1️⃣ What is an Operating System?

Answer:
An Operating System (OS) is system software that acts as an interface between the user and computer hardware.
It manages CPU, memory, files, and devices.

Example: Windows, Linux, macOS.

2️⃣ What are the main functions of OS?

- Process management
- Memory management
- File system management
- Device management
- Security

🔹 Types of Operating System (With Simple Definitions)

1️⃣ Batch Operating System

Definition:
A batch OS executes similar jobs in batches without user interaction.

👉 Users do not interact directly with the system.
👉 Jobs are processed one by one automatically.

Example: Old payroll systems.

3️⃣ Multiprogramming Operating System

Definition:
A multiprogramming OS keeps multiple programs in memory at the same time and switches the CPU among them to increase utilization.

👉 If one process waits, CPU executes another.

4️⃣ Multitasking Operating System

Definition:
A multitasking OS allows a single user to run multiple tasks at the same time.

Example:
Listening to music + browsing + typing in MS Word.

5️⃣ Multiprocessing Operating System

Definition:
A multiprocessing OS uses two or more CPUs to execute processes simultaneously.

👉 Improves performance and reliability.

🔹 1️⃣ Program

Definition:
A program is a set of instructions written in a programming language to perform a specific task.

👉 It is static (stored in disk).
👉 It is not running.

🔹 2️⃣ Process

Definition:
A process is a program in execution.

👉 It is active.
👉 It uses CPU, memory, and other resources.

💡 Short difference:
Program = Passive
Process = Active

🔹 3️⃣ Hard Disk (HD)

Definition:
Hard Disk is a secondary storage device used to permanently store data.

👉 Non-volatile memory
👉 Stores OS, software, files

Example: HDD, SSD

🔹 4️⃣ RAM (Random Access Memory)

Definition:
RAM is primary memory used to temporarily store data and programs currently being used by the CPU.

👉 Volatile memory
👉 Data lost when power off

🔹 5️⃣ ROM (Read Only Memory)

Definition:
ROM is non-volatile memory that permanently stores important instructions like the boot program.

👉 Used to start the computer
👉 Data is not lost when power off

🔹 6️⃣ Cache Memory

Definition:
Cache is a small, high-speed memory located between CPU and RAM that stores frequently used data.

👉 Faster than RAM
👉 Reduces access time

🔹 7️⃣ Register

Definition:
Register is the smallest and fastest memory located inside the CPU.

👉 Stores temporary data and instructions
👉 Very high speed

🔹 1️⃣ Preemptive Scheduling

Definition:
In preemptive scheduling, the CPU can be taken away from a running process before it finishes.

👉 OS can interrupt the process.
👉 Used in time-sharing systems.

Example Algorithms:

- Round Robin
- Shortest Remaining Time First (SRTF)
- Priority (Preemptive)

Example:
If a high-priority process arrives, CPU stops current process and runs the new one.

🔹 2️⃣ Non-Preemptive Scheduling

Definition:
In non-preemptive scheduling, once a process gets the CPU, it runs until it finishes or goes to waiting state.

👉 CPU is not taken away forcefully.
👉 Simple but slower response.

Example Algorithms:

- FCFS
- SJF (Non-preemptive)
- Priority (Non-preemptive)

🔥 Main Differences (Very Important for Viva)

Preemptive	Non-Preemptive
CPU can be taken away	CPU cannot be taken away
Better response time	Poor response time
More complex	Simple
Used in modern OS	Used in simple systems

🔹 1️⃣ Symmetric Multiprocessing (SMP)

Definition:
In Symmetric Multiprocessing, two or more processors share the same memory and are controlled by a single operating system. All processors are equal.

👉 Each processor can perform any task.
👉 Load is distributed among processors.
👉 Processors run in parallel.

Example: Modern multi-core systems.

Short Viva Line:

In SMP, all processors are equal and share the same memory and OS.

🔹 2️⃣ Asymmetric Multiprocessing (AMP)

Definition:
In Asymmetric Multiprocessing, one processor acts as the master and controls the other processors (slaves).

👉 Master processor assigns tasks.
👉 Slave processors execute assigned tasks.
👉 Control is centralized.

Short Viva Line:

In AMP, one processor controls others, and processors are not equal.

🔥 Main Difference (Important for Viva)

Symmetric (SMP)	Asymmetric (AMP)
All processors are equal	One master, others slave
Shared control	Centralized control
Better performance	Less flexible
Used in modern systems	Used in older systems

🔹 **Memory Allocation**

Definition:
Memory allocation is the process of assigning memory space to a program or process.

👉 OS gives required memory to run a program.

🔹 **Memory Deallocation**

Definition:
Memory deallocation is the process of freeing previously allocated memory after it is no longer needed.

👉 OS takes back memory for reuse.

🔹 1️⃣ **Process Management**

Definition:
Process management is the function of the OS that creates, schedules, executes, and terminates processes.

👉 It decides which process gets the CPU.

🔹 2️⃣ **Memory Management**

Definition:
Memory management is the function of the OS that allocates and deallocates memory to processes.

👉 It keeps track of memory usage.

🔹 3️⃣ **File System Management**

Definition:
File system management is the function of the OS that organizes, stores, and retrieves files on storage devices.

👉 It controls file creation, deletion, and access.

🔹 4️⃣ **Device Management**

Definition:
Device management is the function of the OS that controls and manages hardware devices through drivers.

👉 It handles input/output operations.

🔹 1️⃣ **System Call**

Definition:
A system call is a mechanism by which a user program requests services from the operating system.

👉 It provides an interface between user space and kernel space.
👉 Example: read(), write(), fork()

Short Viva Line:

A system call allows a user program to access OS services.

🔹 2️⃣ **User Mode**

Definition:
User mode is the mode in which user applications run with limited access to system resources.

👉 Cannot directly access hardware.
👉 Cannot execute privileged instructions.

Short Viva Line:

User mode runs applications with restricted access.

🔹 3️⃣ Kernel Mode

Definition:
Kernel mode is the privileged mode in which the operating system runs with full access to hardware and memory.

👉 Can execute all instructions.
👉 Controls system resources.

Short Viva Line:

Kernel mode has full access to hardware and system resources.

🔹 Types of System Calls

1️⃣ Process Control

Used to create, execute, and terminate processes.

👉 Examples: fork(), exec(), exit(), wait()

2️⃣ File Management

Used to create, read, write, and delete files.

👉 Examples: open(), read(), write(), close()

3️⃣ Device Management

Used to request and release hardware devices.

👉 Examples: ioctl(), read(), write()

4️⃣ Information Maintenance

Used to get or set system information.

👉 Examples: getpid(), alarm(), sleep()

5️⃣ Communication

Used for inter-process communication (IPC).

👉 Examples: pipe(), shmget(), send(), receive()